

IDS with CNN + LSTM — Journey Keywords & Summary

Internship project on UNSW-NB15 | Aligned with Halbouni et al., IEEE Access 2022

1. What was completed

Phases 1–8: environment (Python 3.12, TensorFlow/Keras), UNSW-NB15 data, EDA notebook, preprocessing (StandardScaler + SelectKBest), CNN / LSTM / hybrid CNN-LSTM models, training with EarlyStopping, evaluation (accuracy, precision, detection rate, F1, FAR, ROC), binary + multiclass results, and academic report. Best binary hybrid result: ~93.9% accuracy, ~94.1% detection rate, ~6.4% FAR, AUC ~0.989 (paper-comparable).

2. Domain & problem

- Intrusion Detection System (IDS)
- Network security
- Cybersecurity
- Anomaly detection
- Signature-based / misuse detection
- Benign vs attack
- Zero-day attacks
- False Alarm Rate (FAR)
- Detection rate
- Network traffic
- Flow-level features

3. Dataset

- UNSW-NB15
- Training / testing set
- Binary classification
- Multiclass classification
- Attack categories (Normal, Fuzzers, DoS, Exploits, Generic, Reconnaissance, Analysis, Backdoor, Shellcode, Worms)
- Class imbalance
- Distribution shift
- Official split
- Stratified split

4. Preprocessing & ML pipeline

- EDA (Exploratory Data Analysis)
- Feature engineering
- Categorical encoding
- LabelEncoder
- Standardization
- StandardScaler
- Feature selection
- SelectKBest
- ANOVA F-score
- Data leakage
- Leakage features
- One-hot encoding
- Train/test split
- Validation set
- Stratified sampling
- Fit on train only

5. Deep learning architecture

- Deep Neural Network (DNN)
- Convolutional Neural Network (CNN)
- Conv1D
- Spatial features
- Feature map
- MaxPooling
- Batch Normalization
- Long Short-Term Memory (LSTM)
- Temporal features
- Sequence learning
- Hybrid CNN-LSTM
- Softmax
- ReLU
- Dropout
- Overfitting
- Dense / fully connected layer
- Adam optimizer
- Categorical cross-entropy
- EarlyStopping
- ReduceLROnPlateau
- Epoch
- Batch size
- Trainable parameters

6. Evaluation metrics

- Confusion matrix
- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)
- Accuracy
- Precision
- Recall / Detection Rate
- F1-score
- False Alarm Rate (FAR)
- ROC curve
- AUC (Area Under Curve)

7. Tools & stack

- Python 3.12
- Virtual environment (venv)
- TensorFlow
- Keras
- pandas
- NumPy
- scikit-learn
- Matplotlib
- Seaborn
- Jupyter
- .keras model file
- Metadata JSON

8. Academic / paper terms

- Halbouni et al.
- IEEE Access 2022
- Benchmark dataset
- Baseline models
- Hybrid deep learning
- Spatial-temporal feature extraction
- Reproducibility
- Internship / academic submission
- Moustafa & Slay (UNSW-NB15)

9. Debugging lessons (viva talking points)

- Validation split bias
- Unshuffled validation
- Silent feature mismatch
- Model–data config mismatch
- Cross-split evaluation error

10. Key results (hybrid CNN-LSTM, stratified)

Binary: Accuracy 93.93% · Precision 96.29% · Detection rate 94.13% · F1 95.20% · FAR 6.42% · AUC 0.989

Multiclass (10 classes): Accuracy 82.15% · Precision 82.60% · Detection rate 82.15% · F1 79.41% · FAR 2.26%

Paper (binary UNSW-NB15): ~93.68% accuracy · ~94.84% detection rate

Reference: Halbouni et al. (2022), IEEE Access, DOI 10.1109/ACCESS.2022.3206425